Spacer composition, not motif grammar, drives regulatory signals in DNA sequence models

Background

The billboard model holds that enhancer activity depends on which binding sites are present—absolute identity—not on their relative positions or spacing [1,2]. This contrarian position challenges the prevailing assumption that motif grammar carries regulatory meaning and that DNA foundation models learn positional syntax. We test the billboard hypothesis directly using a spacer-factored grammar sensitivity framework (SF-GSI) that isolates positional context (motif arrangement) from absolute context (spacer composition)—resolving a confound in standard vocabulary-preserving shuffles, which perturb both simultaneously.

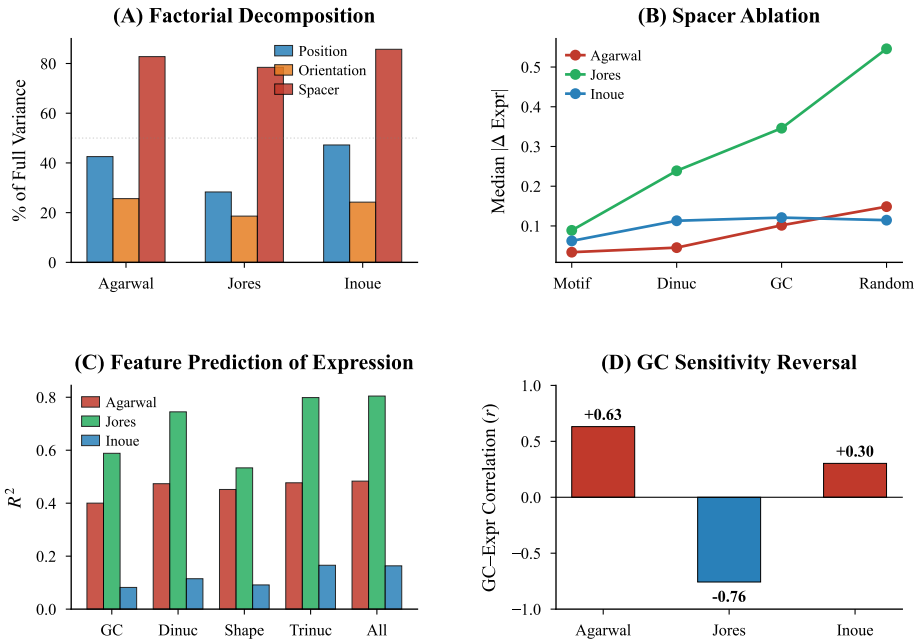

Figure 1: Absolute context dominates apparent grammar effects. Spacer composition (absolute context) explains most measured grammar sensitivity; motif rearrangement (positional context) induces substantially smaller prediction changes.

Methods

SF-GSI computes a grammar sensitivity index, $GSI = \sigma(\hat{y})/|\mu(\hat{y})|$, over vocabulary-preserving motif shuffles and factors the response into positional versus absolute context using matched-spacer perturbations and null-normalized testing. The billboard prediction is concrete: controlling for absolute context should collapse apparent grammar sensitivity to noise. We applied SF-GSI to five MPRA datasets (Agarwal, Klein, Inoue, Vaishnav, and Jores), totaling 7,650 model–enhancer measurements across DNABERT-2, Nucleotide Transformer v2, HyenaDNA, Enformer, and PARM [4–7]. Code: `https://github.com/bryanc5864/GRAMLANG`.

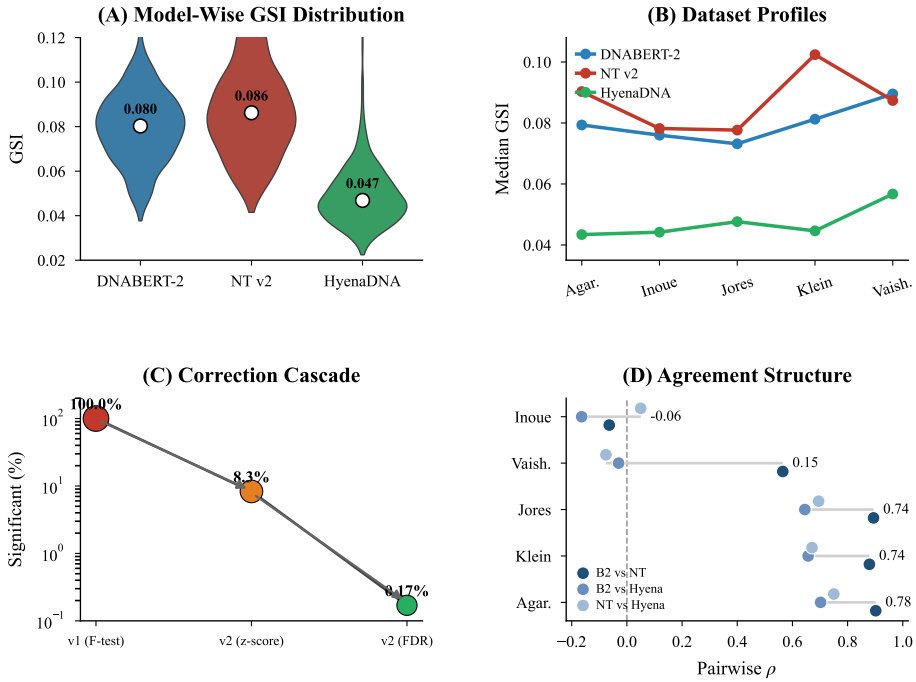

Figure 2: Statistical correction collapses apparent grammar prevalence. Calls fall from 100% under naive thresholding to 8.3% after null normalization and to 0.17% (13/7,650) after FDR correction.

Results

Under naive thresholding all tested enhancers appear grammar-sensitive, but this falls to 8.3% after null normalization and to 0.17% (13/7,650) after FDR correction (Fig. 2). That three-order-of-magnitude collapse is the signature predicted by the billboard model: once absolute context is controlled, apparent positional syntax vanishes. Absolute context (spacer composition) accounts for 78–86% of grammar-sensitivity variance (Fig. 1), while positional context—motif-only rearrangements—produces substantially smaller effects. These models have learned *what* is present, not *where* it is.

Positive controls confirm the framework does not erase genuine grammar: with spacer sequence held fixed, orientation effects remain detectable ($p \leq 7.2 \times 10^{-15}$). Yet these are rare exceptions; variance decompositions assign only 0–1.6% to arrangement versus 8–22% to motif vocabulary, consistent with the billboard prediction. Cross-species transfer is near zero, and cross-model agreement concentrates in specific biological contexts rather than reflecting architecture-general positional learning.

Conclusions

These results provide direct quantitative support for the billboard model: DNA foundation models encode absolute context but show negligible positional context once composition is controlled. Claims of learned regulatory grammar should be made only under perturbation designs that factor out absolute context; without that control, compositional signals masquerade as positional ones.

References: [1] Arnosti & Kulkarni, *J. Cell. Biochem.* 94:890, 2005. [2] Long et al., *Cell* 167:1170, 2016. [3] Castro-Mondragon et al., *Nucleic Acids Res.* 52:D142, 2024. [4] Zhou et al., *bioRxiv* 2023, DNABERT-2. [5] Nguyen et al., *bioRxiv* 2023, HyenaDNA. [6] Avsec et al., *Nature Methods* 18:1196, 2021, Enformer. [7] Agarwal et al., *Science* 380:eadh0559, 2023; Jores et al., *Nat. Plants* 7:842, 2021.